

5291 project

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ii Describe the collected data

Goal

We aim to examine whether the long-term relationship assumed in classical pairs trading remains stable over time and to investigate whether a time-varying hedge ratio (a dynamic beta calculated via the Kalman filter) can enhance the robustness and profitability of the pairs trading strategy.

Pairs

Main pair: **KO–PEP** (similar consumer beverage businesses).

Contrasting pair: **XOM–CVX** (oil majors; potentially more regime-sensitive).

Source and price

Daily **Adjusted Close** from **Yahoo Finance** to account for splits/dividends.

Sample window: 2015-01-01 to 2026-01-01 (daily).

Raw files (CSV)

- KO_2015-01-01_2026-01-01.csv
- PEP_2015-01-01_2026-01-01.csv
- XOM_2015-01-01_2026-01-01.csv
- CVX_2015-01-01_2026-01-01.csv

Each file contains Yahoo columns (Date, Open, High, Low, Close, Adj Close, Volume). We use Date and Adj Close in this project.

Construction

After aligning trading dates and dropping missing observations, let P_t^X, P_t^Y denote adjusted close prices in a pair. We use log prices $x_t = \log(P_t^X)$, $y_t = \log(P_t^Y)$ and log returns $r_t = \Delta \log(P_t)$. The static baseline uses OLS

$$y_t = \alpha + \beta x_t + \varepsilon_t, \quad s_t^{\text{static}} = y_t - \hat{\alpha} - \hat{\beta} x_t.$$

We also compute a rolling z-score (window $W = 252$)

$$z_t = \frac{s_t^{\text{static}} - \mu_t}{\sigma_t},$$

where μ_t, σ_t are rolling mean/std of s_t^{static} .

iii Descriptive statistics

We summarize (a) single-asset daily log returns and (b) pair-level diagnostics (static baseline).

Ticker	Mean	Std	Skew	Kurt	Min	Max
KO	-0.000309	0.011228	0.821762	10.343775	-0.062702	0.101709
PEP	-0.000271	0.012008	0.448434	18.800564	-0.121689	0.121322
XOM	-0.000265	0.017284	0.188605	6.456925	-0.119310	0.130511
CVX	-0.000280	0.018466	0.966415	25.795047	-0.205015	0.249906

Table 1: Daily log return summary (2015-01-01 to 2026-01-01).

Pair	$\text{Corr}(r^X, r^Y)$	$\hat{\beta}$	$\text{Std}(s^{\text{static}})$	% $ z > 2$
KO-PEP	0.725794	0.868284	0.095088	7.881
XOM-CVX	0.833666	0.963535	0.17081	3.651

Table 2: Pair-level diagnostics (static baseline).

iv Graphics

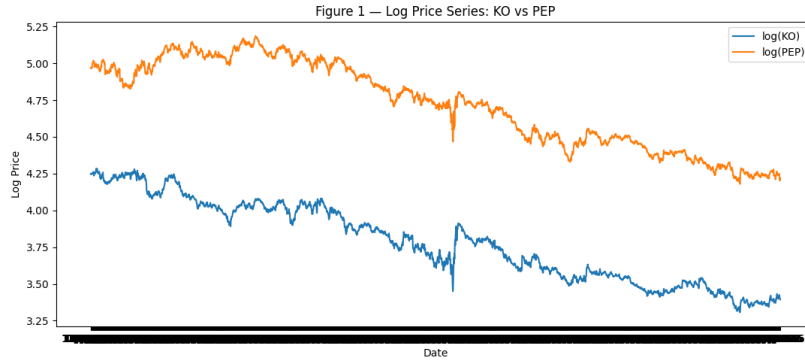


Figure 1: KO vs PEP: log adjusted close prices

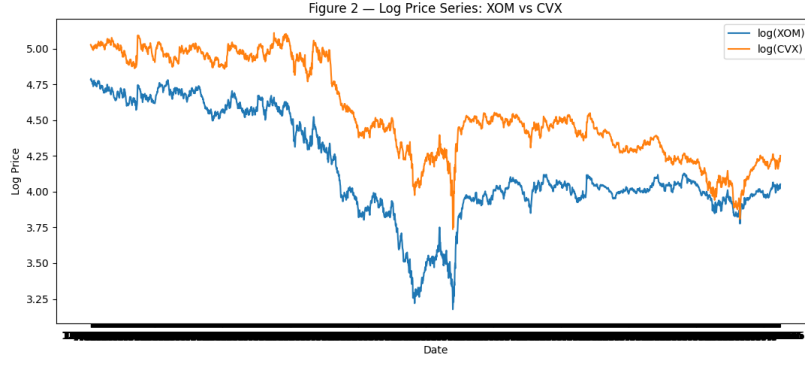


Figure 2: XOM vs CVX: log adjusted close prices.

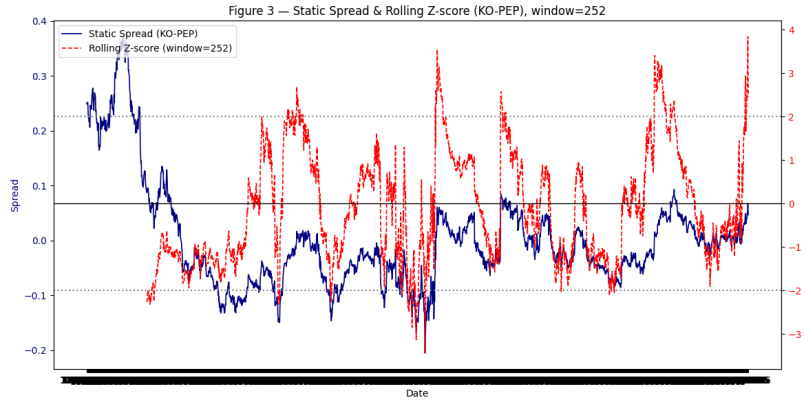


Figure 3: KO-PEP: static spread and rolling z-score ($W = 252$).

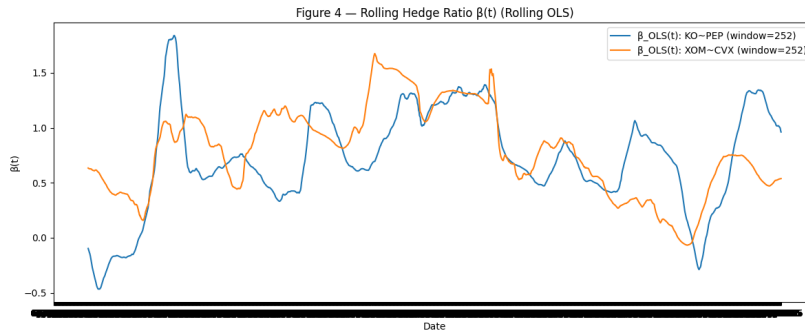


Figure 4: Rolling OLS hedge ratio $\hat{\beta}_{\text{roll}}(t)$ ($W = 252$) for both pairs.

Figure 5 — 3D: Time vs Z-score vs Rolling ADF p-value

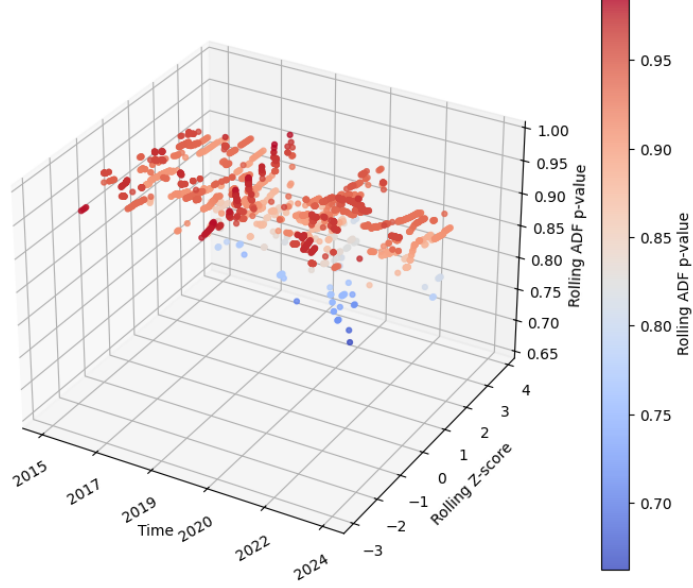


Figure 5: 3D plot: time (x), z-score (y), color = rolling ADF p-value of the spread.

v Comments

Table 1 shows clear cross-sector differences: KO/PEP have lower daily volatility than XOM/CVX, and all four return series exhibit strong fat tails (large excess kurtosis), implying that extreme moves are non-negligible and spread signals may be less stable under shocks.

Table 2 summarizes the static baseline. Both pairs have strong return co-movement, with XOM–CVX more highly correlated than KO–PEP. However, the static spread is much more volatile for XOM–CVX, while KO–PEP exhibits a higher frequency of large standardized spread deviations ($|z| > 2$) under our normalization. Overall, high correlation alone does not guarantee a stable residual relationship under a fixed hedge ratio, motivating a time-varying hedge ratio model (state-space / Kalman filter) in the final project.

Figures 1–2 show strong co-movement within both pairs, but also clear divergence episodes (more pronounced for XOM–CVX), suggesting the long-run relationship may be regime-dependent. Figure 3 shows that during the observation period for KO–PEP, the standardized spread signal (rolling z-score) exhibits clustered extreme values and time-varying amplitude, rather than a stable mean-reverting pattern. Figure 4 further indicates that the hedge ratio is not constant: rolling OLS estimates drift materially for both pairs, so a fixed- β specification can be misspecified in some periods and may itself in-

duce spread instability. Figure 5 links spread deviations to rolling stationarity diagnostics: periods with large $|z|$ tend to coincide with weaker evidence of stationarity (higher rolling ADF p-values), reinforcing the hypothesis that “tradability” varies over time. Overall, the graphics motivate our final-project extension to a time-varying hedge ratio model (state-space / Kalman filter) and a regime-aware evaluation of spread stability.

vi Research Questions

This project addresses four key research questions:

1. whether the selected stock pairs exhibit statistically significant and stable cointegration;
2. whether the constant hedge ratio assumption underlying traditional pairs trading holds over time;
3. whether incorporating a time-varying hedge ratio via a Kalman filter improves spread stationarity and trading robustness; and
4. whether mean reversion strength is regime-dependent across different market conditions.

The objective of this project is to conduct a comprehensive statistical evaluation of pairs trading strategies by integrating cointegration analysis with time-varying parameter modeling. First, we aim to test whether selected stock pairs within the same industry (KO–PEP and XOM–CVX) exhibit a stable long-run equilibrium relationship using Engle–Granger cointegration tests. We then investigate whether this equilibrium relationship remains constant over time or experiences structural drift across different market regimes.

A key research question is whether the traditional assumption of a constant hedge ratio is appropriate. To address this, we compare a static OLS-based hedge ratio with a time-varying hedge ratio estimated via a Kalman filter in a state-space framework. We examine how dynamic hedge ratio estimation affects spread stationarity, z-score behavior, and the persistence of deviations from equilibrium.

Additionally, we evaluate whether mean reversion strength is regime-dependent by analyzing rolling ADF p-values and rolling hedge ratios. This allows us to assess whether periods of weakened stationarity correspond to prolonged spread deviations and reduced tradability.

Finally, we compare the performance of static versus dynamic trading rules based on spread z-scores, including entry/exit behavior and sensitivity to structural shifts. Overall, the project seeks to determine whether incorporating time-varying parameters enhances the robustness and adaptability of statistical arbitrage strategies under changing market conditions.

1.ii Selected Research Paper

1. Kim, K. (2011). *Performance Analysis of Pairs Trading Strategy Utilizing High Frequency Data with an Application to KOSPI 100 Equities*.
2. Zhang, G. (2020). Pairs Trading with Nonlinear and Non-Gaussian State Space Models. arXiv preprint arXiv:2005.09794.
3. Chen, C., & Alexiou, C. (2025). Cointegration-based pairs trading: Identifying and exploiting similar exchange-traded funds.

1.iii Paper Summary

1. Pairs trading is a well-known statistical arbitrage strategy in financial markets. The basic idea is that two related stocks usually move together over time. When the price difference between the two stocks becomes unusually large, traders expect the spread to return to its normal level. Therefore, investors can open opposite positions in the two stocks and attempt to profit when the prices converge again.

This paper studies how pairs trading works when high-frequency data is used. The author focuses on stocks from the KOSPI 100 index in the Korean stock market. The goal of the paper is to examine whether pairs trading can generate profits when trades are executed over very short time intervals.

The first step in the strategy is selecting stock pairs. The paper chooses pairs based on historical correlation. Stocks that show strong relationships in past price movements are selected as potential trading pairs. After the pairs are chosen, the spread between the two stock prices is calculated and monitored during the trading period.

When the spread becomes much larger or smaller than its historical average, the strategy opens positions. The trader buys the undervalued stock and sells the overvalued stock at the same time. When the spread moves back toward the historical mean, the positions are closed and the profit is realized.

The paper also examines the role of transaction costs and trading frequency. High-frequency data allows the strategy to identify more trading opportunities. However, frequent trading also increases transaction costs. The empirical results show that the strategy can generate positive returns before transaction costs are considered. After realistic transaction costs are included, the profitability becomes much lower.

Many studies have tried to improve pairs trading strategies using different statistical and machine learning methods. Some research uses time-varying hedge ratios to better describe the relationship between assets. Other studies apply state space models and Kalman filtering to update

model parameters over time. More recent research also combines machine learning techniques with traditional statistical models to improve pair selection and trading signals.

Overall, this paper provides an empirical analysis of pairs trading using high-frequency data from the Korean stock market. It explains how the strategy works in practice and shows the impact of transaction costs on trading performance.

2. This paper proposes a new framework for improving the performance of pairs trading, a popular statistical arbitrage strategy widely used in hedge funds and quantitative trading. Pairs trading involves identifying two historically correlated securities and constructing a market-neutral portfolio by going long one asset and short the other when their price relationship deviates from its long-term equilibrium. Profits are generated when the spread between the assets reverts to its historical mean.

The study argues that many traditional pairs trading models are overly simplified because they assume that the spread between assets follows linear, Gaussian, and homoscedastic dynamics. However, real financial time series often exhibit nonlinear behavior, heavy-tailed distributions, and volatility clustering, which cannot be adequately captured by conventional models. To address this limitation, the author introduces a nonlinear and non-Gaussian state-space model in which the spread between two assets is treated as an unobservable latent variable that follows a mean-reverting process.

The proposed model has three key features. First, it allows the innovations of the spread to be non-Gaussian, enabling the model to capture heavy-tailed distributions observed in financial markets. Second, the model incorporates heteroskedasticity, allowing volatility to change over time and reflect volatility clustering. Third, the mean-reversion dynamics of the spread can be nonlinear, providing greater flexibility in modeling financial asset relationships. These improvements allow the model to represent more realistic market dynamics compared to existing pairs trading frameworks.

To estimate the unobservable spread and model parameters, the paper applies a Quasi-Monte Carlo Kalman Filter (QMCKF). This filtering method extends the traditional Kalman filter to accommodate nonlinear and non-Gaussian state-space models. Once the spread is estimated, it becomes the primary signal used to generate trading decisions.

The paper also proposes a new trading strategy (Strategy C) designed to improve risk management and profitability compared with traditional strategies. Conventional strategies typically open a trade when the spread crosses predetermined upper or lower thresholds and close the position when the spread returns to its mean. In contrast, Strategy C modifies the entry and exit rules to reduce risk exposure and adapt to changing volatility conditions. In addition, the paper introduces a simulation-based

Monte Carlo method for determining optimal trading thresholds, allowing the strategy to maximize performance measures such as expected return or Sharpe ratio.

Empirical analysis is conducted using several datasets, including Pepsi vs. Coca-Cola (PEP/KO) and Taiwan vs. Hong Kong ETFs (EWT/EWH). The results show that the proposed nonlinear state-space model combined with Strategy C significantly improves trading performance. For example, the model achieves an annualized return of 21.86% for the PEP/KO pair and 31.84% for the EWT/EWH pair, outperforming conventional approaches in both returns and Sharpe ratios.

Further tests on pairs of U.S. banks listed on the NYSE confirm the robustness of the proposed approach. The results demonstrate consistent improvements in both returns and risk-adjusted performance, particularly when the spread exhibits heteroskedastic volatility.

In conclusion, the paper shows that incorporating nonlinear dynamics, non-Gaussian noise, and heteroskedastic volatility into the modeling of the spread can significantly enhance the effectiveness of pairs trading strategies. The findings highlight the importance of realistic financial modeling and adaptive trading rules for improving the profitability and robustness of statistical arbitrage strategies.

3. Chen and Alexiou (2025) study the effectiveness of cointegration-based pairs trading in the ETF market. Using data from 2000 to 2024, the paper examines 30 ETF pairs across different asset classes and sectors to evaluate how cointegration stability affects profitability and risk. In particular, it considers whether lower z -score thresholds improve trading performance and whether ETF pairs exhibit sufficiently stable long-run relationships to support mean reversion.

The paper is motivated by the fact that ETFs differ from ordinary stocks in important ways, including tracking error, fund composition, and liquidity mechanisms. For this reason, conclusions from the traditional stock-based pairs trading literature may not apply directly to ETFs. The paper therefore argues that similar historical price movements alone are not enough to identify tradable pairs; instead, what matters is whether two ETFs maintain a statistically meaningful long-run equilibrium relationship.

Methodologically, the paper implements a cointegration-based strategy using ETF price ratios and stationarity tests. It identifies potentially tradable pairs and generates trading signals from z -scores based on the ratio relationship. The results show that lower z -score thresholds create more trading opportunities and may improve profits and Sharpe ratios, but they also lead to higher volatility and deeper drawdowns.

A central finding is that the success of ETF pairs trading depends critically on the persistence of cointegration. The paper shows that highly similar price paths do not necessarily imply a stable tradable relationship. For

example, SPY and IVV appear relatively stable, whereas QQQ and XLK, despite strong correlation, do not exhibit persistent stationarity over the full sample. More broadly, many ETF pairs remain cointegrated only for short periods, which limits long-run profitability.

This paper is directly relevant to our project because it highlights a key limitation of conventional pairs trading: the long-run relationship between paired assets is often unstable. Our project similarly asks whether the assumption of a constant hedge ratio is too restrictive and whether a time-varying hedge ratio estimated through a Kalman filter can improve spread stationarity and trading robustness. In this sense, the paper provides a clear motivation for extending a static framework to a dynamic one.

1.v Introduction

Pairs trading is a classic market-neutral strategy based on the idea that two related assets may share a long-run equilibrium relationship. When the spread between them deviates from its historical norm, traders take offsetting positions and expect the spread to revert to the mean. Because this strategy relies on spread stationarity and mean reversion, it is commonly viewed as a form of statistical arbitrage.

Previous studies show that the profitability of pairs trading depends heavily on how the relationship between paired assets is modeled. Kim (2011) studies pairs trading using high-frequency data from KOSPI 100 equities and finds that, although the strategy can create arbitrage opportunities, its performance is sensitive to transaction costs and trading frequency. More recent research has challenged the assumption that the relationship between paired assets remains constant over time. Zhang (2020) proposes a state-space framework that allows spread dynamics to vary over time, while Chen and Alexiou (2025) show that even when two assets exhibit similar historical price movements, their cointegrating relationship may not remain stable over the full sample period. Together, these studies highlight an important limitation of static pairs trading models.

Motivated by this literature, our project investigates whether the assumption of a constant hedge ratio is too restrictive in practice. Specifically, we examine whether a time-varying hedge ratio estimated through a Kalman filter can better capture the evolving relationship between paired assets and improve the robustness of trading signals. We then apply the Augmented Dickey–Fuller test to evaluate spread stationarity and use the Ornstein–Uhlenbeck process to assess mean reversion strength. Our goal is to determine whether a dynamic hedge-ratio framework provides a more reliable and profitable approach than the traditional static specification.

1.ii Preliminary Models and Summary Output

The goal of this project is to study whether the long-run relationship assumed in classical pairs trading remains stable over time, and whether a time-varying hedge ratio may eventually improve robustness relative to a static model.

We fit two models to study the relationship within each stock pair.

Model 1

As a baseline specification for this assignment, a static linear regression model was fitted for each pair using daily adjusted closing prices from Yahoo Finance over the sample period 2015-01-01 to 2026-01-01.

The main pair is KO–PEP, and the contrasting pair is XOM–CVX. For each pair, the following OLS model was estimated:

$$P_{1,t} = \alpha + \beta P_{2,t} + \varepsilon_t,$$

where $P_{1,t}$ and $P_{2,t}$ denote the adjusted closing prices of the two assets at time t , β is interpreted as the constant hedge ratio, and the residual ε_t is treated as the spread.

For KO-PEP, the fitted baseline model is

$$\widehat{KO}_t = 5.8701 + 0.3396 PEP_t.$$

The slope coefficient is highly significant ($p < 2 \times 10^{-16}$), with $R^2 = 0.8106$ and adjusted $R^2 = 0.8105$. The residual standard error is 5.422. These results indicate a strong linear comovement in prices under the static specification.

For XOM-CVX, the fitted baseline model is

$$\widehat{XOM}_t = -0.8515 + 0.6863 CVX_t.$$

The slope coefficient is again highly significant ($p < 2 \times 10^{-16}$), with $R^2 = 0.8556$ and adjusted $R^2 = 0.8556$. The residual standard error is 9.571. This pair also exhibits strong price comovement under a constant hedge-ratio model.

Table 3 summarizes the main regression results.

Table 3: Baseline OLS summary for the two pairs

Pair	Intercept	Slope (β)	Slope p -value	R^2	Adj. R^2	RSE
KO-PEP	5.8701	0.3396	$< 2 \times 10^{-16}$	0.8106	0.8105	5.422
XOM-CVX	-0.8515	0.6863	$< 2 \times 10^{-16}$	0.8556	0.8556	9.571

Model 2

In addition to the static OLS baseline, we also fitted a dynamic hedge-ratio model in order to allow the relationship between the two assets to vary over time. Specifically, for each pair we modeled log prices using a state-space specification:

$$\log P_{1,t} = \alpha_t + \beta_t \log P_{2,t} + \varepsilon_t,$$

where both the intercept α_t and the hedge ratio β_t are allowed to evolve over time and are estimated recursively using a Kalman-filter style update. Under this model, the dynamic spread is defined as

$$s_t^{dyn} = \log P_{1,t} - \hat{\alpha}_t - \hat{\beta}_t \log P_{2,t}.$$

For KO-PEP, the estimated dynamic hedge ratio had mean 0.8789, standard deviation 0.0982, minimum 0.7172, and maximum 1.1472. For XOM-CVX, the estimated dynamic hedge ratio had mean 0.9530, standard deviation 0.0713, minimum 0.8283, and maximum 1.0958. These results suggest that the hedge ratio is not constant over time for either pair.

Table 4 summarizes the dynamic-model hedge-ratio estimates.

Table 4: Dynamic hedge-ratio summary from the state-space model

Pair	Mean β_t	SD β_t	Min β_t	Max β_t	Dynamic Spread SD
KO-PEP	0.8789	0.0982	0.7172	1.1472	0.0026
XOM-CVX	0.9530	0.0713	0.8283	1.0958	0.0032

Taken together, the static model provides a simple and interpretable benchmark, while the dynamic model allows the hedge ratio to adjust over time and is more closely aligned with the main objective of the final project.

1.iii Testing a Research Question

A central research question is whether a constant hedge ratio is sufficient to produce a stable spread for pairs trading. To examine this, the residual spread from each OLS model was tested for stationarity using the Augmented Dickey–Fuller (ADF) test. Specifically, this two-step procedure follows the Engle-Granger methodology for cointegration testing. By regressing one price series on the other and testing the residuals for a unit root, we evaluate whether the assets share a stable, long-term equilibrium under a fixed hedge ratio.

For KO–PEP, the ADF test on the full-sample spread gives a test statistic of -1.2112 with $p = 0.9047$. For XOM–CVX, the corresponding statistic is -1.6917 with $p = 0.7089$. In both cases, the null hypothesis of a unit root cannot be rejected.

Thus, although the static OLS baseline captures price comovement well, the residual spreads do not appear stationary under this specification. This suggests that a constant hedge ratio may be too restrictive for both pairs and provides motivation for extending the analysis to a time-varying hedge-ratio framework in the final project.

Table 5 reports the full-sample ADF results.

Table 5: ADF test results for full-sample residual spreads

Pair	Dickey–Fuller Statistic	Lag Order	p -value
KO–PEP	-1.2112	14	0.9047
XOM–CVX	-1.6917	14	0.7089

The dynamic model provides additional evidence against the constant-hedge-ratio assumption. In particular, the estimated β_t series varies meaningfully over time for both KO–PEP and XOM–CVX rather than remaining close to a fixed constant. This suggests that the relationship between the two stocks evolves over time and that a static spread may be misspecified even if the two price series exhibit strong visual comovement.

Therefore, the combined evidence from the ADF test and the dynamic hedge-ratio estimates supports the conclusion that a constant hedge ratio is likely too restrictive for these two pairs.

1.iv Model Validation

Model 1

A simple train-test validation procedure was used to assess stability. Each sample was divided into 80% training observations and 20% testing observations. The OLS model was estimated on the training set, and the fitted coefficients were then used to construct out-of-sample spreads on the test set.

For KO-PEP, the training-sample model is

$$\widehat{KO}_t = 9.7925 + 0.2884 PEP_t.$$

The training-sample spread has standard deviation 1.8659, while the test-sample spread has standard deviation 7.7883. The ADF test on the test-sample spread yields a statistic of -1.6701 with $p = 0.7179$, again failing to reject the unit-root null. This suggests weak out-of-sample stability under the static model.

For XOM-CVX, the training-sample model is

$$\widehat{XOM}_t = 9.2849 + 0.5504 CVX_t.$$

The training-sample spread has standard deviation 8.4313, and the test-sample spread has standard deviation 5.0373. However, the ADF test on the test-sample spread gives a statistic of -1.9056 with $p = 0.6183$, which still fails to reject the unit-root null.

Overall, the validation results reinforce the same conclusion: the static regression captures a substantial amount of price comovement, but the implied spreads are not sufficiently stable to support a strong mean-reverting interpretation. This is especially clear for KO-PEP, where the spread volatility increases sharply out of sample.

Given the failure to reject the unit-root null in both pairs, the static OLS approach appears inadequate for a robust pairs trading strategy. Consequently, the final project will transition to a State-Space Model using the Kalman Filter. This will allow the hedge ratio to evolve as a latent stochastic process, potentially restoring stationarity to the resulting spread.

Table 6 summarizes the validation results.

Table 6: Train-test validation summary

Pair	Train Spread SD	Test Spread SD	Test ADF Statistic	Test p -value
KO-PEP	1.8659	7.7883	-1.6701	0.7179
XOM-CVX	8.4313	5.0373	-1.9056	0.6183

Model 2

We also validated the dynamic model using the time path of the estimated hedge ratio and one-step-ahead forecast errors from the recursive state-space specification. Table 7 shows that the estimated hedge ratio varies materially over time for both pairs. For KO-PEP, β_t ranges from 0.7172 to 1.1472, while for XOM-CVX it ranges from 0.8283 to 1.0958. The one-step-ahead forecast errors are also small on the log-price scale, with RMSE equal to 0.0087 for KO-PEP and 0.0108 for XOM-CVX. These results suggest that the dynamic model adapts to gradual changes in the pair relationship over time.

Table 7: Dynamic-model validation summary

Pair	Mean β_t	SD β_t	Min β_t	Max β_t	RMSE	MAE
KO-PEP	0.8789	0.0982	0.7172	1.1472	0.0087	0.0059
XOM-CVX	0.9530	0.0713	0.8283	1.0958	0.0108	0.0069

We also examined one-step-ahead forecast errors from the dynamic model. For most of the sample, these forecast errors remain relatively close to zero, indicating that the recursive specification is able to adapt to changes in the pair relationship over time. This behavior is consistent with the very small dynamic-spread standard deviations reported in Table 4.

Overall, the baseline validation shows that a fixed-coefficient spread is not stably mean reverting, while the dynamic-model validation shows that the hedge ratio itself changes over time and can be tracked by a recursive specification. These findings support the use of the dynamic model as a more flexible extension of the baseline benchmark.

1.v Graphics

Figures 6a–6d show the price paths and OLS residual spreads for the two pairs.

For KO-PEP, the two price series comove closely, but the residual spread drifts upward in the later part of the sample and does not remain centered around zero. For XOM-CVX, the price series also move together overall, yet the spread shows substantial drift and regime-like variation, especially around 2020 and the subsequent recovery period.

In conclusion, the graphics are consistent with the ADF results and suggest that the static OLS model captures price comovement but does not produce sufficiently stable residual spreads for either pair.



(a) Adjusted closing prices for KO and PEP.



(b) Residual spread from the static OLS model for KO-PEP.



(c) Adjusted closing prices for XOM and CVX.



(d) Residual spread from the static OLS model for XOM-CVX.

Figure 6: Price paths and residual spreads for KO-PEP and XOM-CVX under the static OLS specification.

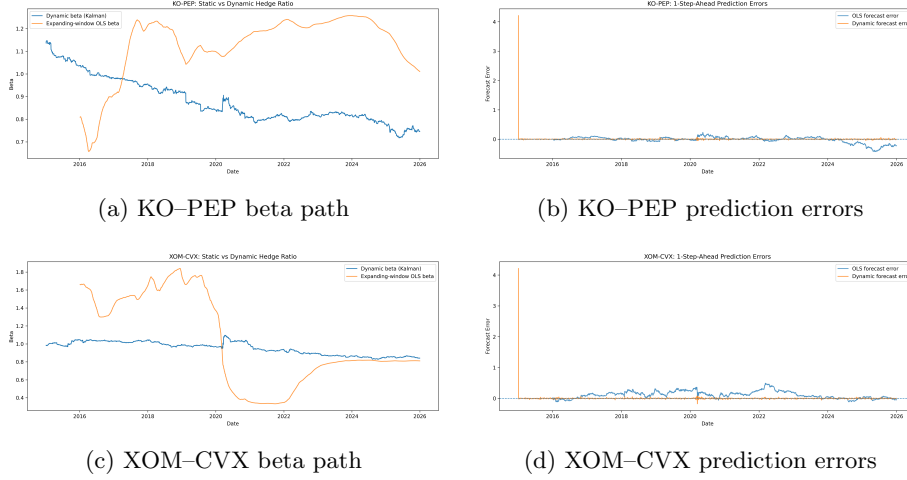


Figure 7: Dynamic-model diagnostics for the two stock pairs. The beta-path plots show that the estimated hedge ratio varies over time for both KO-PEP and XOM-CVX. The prediction-error plots show that, aside from the initial burn-in period of recursive estimation, forecast errors remain relatively small and close to zero for most of the sample.

The dynamic-model figures provide additional support for this conclusion. The time-varying beta plots show that the estimated hedge ratio changes substantially over time for both pairs rather than remaining fixed. The one-step-ahead prediction error plots show that, after the initial burn-in period of recursive estimation, forecast errors remain relatively small and close to zero for most of the sample, suggesting that the dynamic model is better able to adapt to changes in the relationship between the two assets.

Taken together, the graphics support the statistical conclusion from Parts 1.iii and 1.iv, that the baseline OLS model explains price comovement but does not generate sufficiently stable residual spreads, while the dynamic model provides evidence that the hedge ratio itself varies over time.

1.iv Bibliography

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